

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 8: JOURNEY INSIDE THE ATOM [SET B - ADVANCED HOTS]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH08/SET-B****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET B - HIGHER ORDER THINKING SKILLS):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Advanced Conceptual and Numerical MCQs with single correct option.
3. **Section B (Q16 – Q20):** Critical Assertion-Reasoning based questions testing analytical clarity.
4. **Section C (Q21 – Q25):** Case-Study / Practical Experimental Data-interpretation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: ADVANCED & NUMERICAL MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 Rutherford deduced the nuclear radius of an atom to be approximately 10^{-15} m, whereas the atomic radius is about 10^{-10} m. What is the approximate ratio of atomic volume to nuclear volume?

- (A) 10^5 (B) 10^{15}
(C) 10^{10} (D) 10^3

Q.2 Why did J.J. Thomson find that the charge-to-mass ratio (e/m) of cathode rays was constant regardless of the gas used in the tube, whereas canal ray e/m varied with the gas?

- (A) Cathode rays are composed of universal fundamental electrons, whereas canal rays consist of positive gaseous ions whose masses depend on the specific gas element
(B) Canal rays contain neutrons
(C) Cathode rays are electromagnetic waves
(D) Canal rays travel at the speed of light

Q.3 Naturally occurring magnesium consists of three isotopes: ^{24}Mg (79%), ^{25}Mg (10%), and ^{26}Mg (11%). The average atomic mass of magnesium is:

- (A) 24.32 u (B) 25.00 u
(C) 24.00 u (D) 24.65 u

Q.4 Why does potassium ($Z = 19$) have an electronic configuration of 2, 8, 8, 1 rather than 2, 8, 9, even though the M-shell can theoretically accommodate up to 18 electrons?

- (A) The octet rule dictates that the outermost valence shell cannot exceed 8 electrons, so the 19th electron enters the 4th (N) shell
(B) The M-shell can only ever hold 8 electrons
(C) Potassium loses an electron immediately
(D) Neutrons prevent filling the M-shell

Q.5 An atom has 17 protons, 18 neutrons, and 18 electrons. This species is:

- (A) A neutral Argon atom ($^{35}_{18}\text{Ar}$) (B) A chloride anion ($^{35}_{17}\text{Cl}^-$)
(C) A potassium cation ($^{39}_{19}\text{K}^+$) (D) A radioactive chlorine isotope

Q.6 Which pair represents isobars (same mass number, different atomic numbers)?

- (A) $^{12}_6\text{C}$ and $^{14}_6\text{C}$ (B) $^{40}_{18}\text{Ar}$ and $^{40}_{20}\text{Ca}$
 (C) ^1_1H and ^2_1H (D) $^{16}_8\text{O}$ and $^{18}_8\text{O}$

Q.7 How many neutrons are present in the nucleus of a tritium isotope (^3_1H)?

- (A) 1 (B) 2
 (C) 3 (D) 0

Q.8 Which subatomic particle was discovered by James Chadwick in 1932 by bombarding a thin sheet of beryllium with alpha particles?

- (A) Positron (B) Neutron
 (C) Proton (D) Neutrino

Q.9 The electronic configurations of four elements are: P (2, 8, 1), Q (2, 8, 7), R (2, 8, 8), and S (2, 6). Which element exhibits a valency of 2?

- (A) Element P (B) Element Q
 (C) Element R (D) Element S

Q.10 Which isotope is used as standard fuel in nuclear fission reactors to generate electricity?

- (A) Uranium-235 ($^{235}_{92}\text{U}$) (B) Carbon-14 ($^{14}_6\text{C}$)
 (C) Cobalt-60 ($^{60}_{27}\text{Co}$) (D) Iodine-131 ($^{131}_{53}\text{I}$)

Q.11 What is the maximum number of electrons that can be accommodated in the N-shell ($n = 4$)?

- (A) 8 (B) 18
 (C) 32 (D) 50

Q.12 Which of the following ions is isoelectronic with the noble gas Neon ($Z = 10$, configuration 2, 8)?

- (A) Na^+ ($Z=11$) (B) Mg^{2+} ($Z=12$)
 (C) O^{2-} ($Z=8$) (D) All of the above

Q.13 An element X has valency 3 and element Y has valency 2. The chemical formula of the compound formed between X and Y is:

- (A) XY (B) X_2Y_3
 (C) X_3Y_2 (D) X_2Y

Q.14 What is the mass of one mole of neutrons approximately, given the mass of a single neutron is 1.675×10^{-24} g and Avogadro's constant is 6.022×10^{23} ?

- (A) 1.008 g (B) 1840 g
 (C) 0.001 g (D) 100 g

Q.15 Which radioactive isotope is utilized in medicine for the assessment and treatment of thyroid gland disorders?

- (A) Iodine-131 ($^{131}_{53}\text{I}$) (B) Cobalt-60
 (C) Carbon-14 (D) Phosphorus-32

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).

- (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): An atom remains stable and its electrons do not spiral into the nucleus.

Reason (R): According to Neils Bohr, electrons revolve only in discrete stationary orbits without radiating electromagnetic energy.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): Isotopes of chlorine, $^{35}_{17}\text{Cl}$ and $^{37}_{17}\text{Cl}$, have identical chemical properties.

Reason (R): Chemical properties are governed by valence electrons, and both isotopes have identical atomic number 17 and electronic configuration (2, 8, 7).

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): The atomic mass of an element is often a fractional decimal number (e.g., Chlorine is 35.5 u).

Reason (R): Naturally occurring elements often exist as mixtures of two or more isotopes with different natural abundance percentages.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Noble gases (Helium, Argon, Neon) have zero valency.

Reason (R): Their outermost shells are completely filled, leaving no tendency to gain, lose, or share electrons.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): Thomson's model predicted that alpha particles would easily pass through gold atoms with only minimal deflection.

Reason (R): In Thomson's model, positive charge and mass were thought to be spread thinly throughout the entire volume of the atom.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL DATA & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Mass Spectrometry & Nuclear Nucleon Identification

A mass spectrometry laboratory analyzes a radioactive mineral sample containing four nuclear species: P, Q, R, and S. The laboratory records the following data:

- **Species P:** 6 protons, 6 neutrons, 6 electrons
- **Species Q:** 6 protons, 8 neutrons, 6 electrons
- **Species R:** 7 protons, 7 neutrons, 7 electrons
- **Species S:** 8 protons, 8 neutrons, 10 electrons

Q.21 What is the mass number (A) of Species Q?

- (A) 12 (B) 14
(C) 6 (D) 8

Q.22 Which two species are isotopes of each other?

- (A) P and Q (B) Q and R
(C) R and S (D) P and S

Q.23 Which two species have the same mass number ($A = 14$) and are therefore isobars?

- (A) P and Q (B) Q and R
(C) P and R (D) R and S

Q.24 What is the net electrical charge and identity of Species S?

- (A) Neutral oxygen atom (O) (B) Oxide anion with -2 charge (O^{2-})
(C) Nitrogen cation with $+2$ charge (D) Neutral Neon atom

Q.25 Species Q ($^{14}_6\text{C}$) is famously utilized in archeology for which scientific dating technique?

- (A) Radiocarbon dating of biological organic artifacts (B) Uranium-lead dating of volcanic rocks
(C) Potassium-argon dating of meteorites (D) Thermoluminescence of pottery

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH EXPLANATIONS (SET B)

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.1	(B)	$V \propto r^3$. Ratio $= \left(\frac{10^{-10}}{10^{-15}}\right)^3 = (10^5)^3 = 10^{15}$. The atom is mostly empty space.
Q.2	(A)	Electrons are identical in all matter, giving fixed e/m . Positive canal ray particles are stripped residual gas cations; heavier gases produce heavier ions with lower e/m .
Q.3	(A)	$\text{Mass} = \frac{(24 \times 79) + (25 \times 10) + (26 \times 11)}{100} = \frac{1896 + 250 + 286}{100} = \frac{2432}{100} = 24.32 \text{ u}$.
Q.4	(A)	According to the Bohr-Bury octet rule, the outermost valence shell cannot hold more than 8 electrons; hence, the 19th electron starts the N-shell (4s level).
Q.5	(B)	$Z = 17$ (Chlorine). Mass number $A = 17 + 18 = 35$. With 18 electrons (one extra than 17 protons), it carries a net -1 charge (${}^{35}_{17}\text{Cl}^-$).
Q.6	(B)	Argon ($Z=18$) and Calcium ($Z=20$) both have mass number 40; they are isobars.
Q.7	(B)	$\text{Neutrons} = A - Z = 3 - 1 = 2$.
Q.8	(B)	James Chadwick observed penetrating neutral radiation consisting of uncharged particles with mass equal to a proton: neutrons (${}^1_0\text{n}$).
Q.9	(D)	Element S has 6 valence electrons; to complete its octet, it needs $8 - 6 = 2$ electrons. Valency $= 2$ (Oxygen).
Q.10	(A)	Uranium-235 undergoes controlled thermal neutron-induced nuclear fission inside commercial nuclear power reactors.
Q.11	(C)	Maximum capacity $= 2n^2 = 2(4^2) = 2(16) = 32$ electrons.
Q.12	(D)	All three have 10 electrons ($11-1=10$, $12-2=10$, $8+2=10$), matching Neon's 2, 8 configuration.
Q.13	(B)	Criss-cross valency method: X^3 and Y^2 yields X_2Y_3 .
Q.14	(A)	$1.675 \times 10^{-24} \times 6.022 \times 10^{23} \approx 1.008 \text{ g}$.
Q.15	(A)	The thyroid gland absorbs iodine; radioactive Iodine-131 is used to image and treat goiter and thyroid carcinoma.
Q.16	(A)	Bohr postulated quantized non-radiating orbits, overcoming the fatal flaw in Rutherford's model where accelerated charges radiate energy.
Q.17	(A)	Chemical reactivity depends strictly on outer valence electron configuration, which is identical in both isotopes.
Q.18	(A)	Average atomic mass is a weighted average of isotopic masses based on natural fractional abundances (e.g. $75\% \text{ }^{35}\text{Cl} + 25\% \text{ }^{37}\text{Cl} = 35.5 \text{ u}$).
Q.19	(A)	A complete duplet (He) or octet (Ne, Ar) imparts exceptional thermodynamic stability, resulting in zero chemical valency.
Q.20	(A)	Rutherford expected minor deflections because Thomson's diffuse positive sphere could not exert sufficient concentrated electric field to deflect fast alpha particles backward.
Q.21	(B)	$A = \text{protons} + \text{neutrons} = 6 + 8 = 14$.
Q.22	(A)	Species P and Q both have $Z = 6$ (protons), but different neutron counts (6 vs 8); they are isotopes of Carbon (${}^{12}_6\text{C}$ and ${}^{14}_6\text{C}$).
Q.23	(B)	Species Q ($A = 6+8=14$) and Species R ($A = 7+7=14$) have different atomic numbers (6 and 7) but the same mass number (14); they are isobars.
Q.24	(B)	$Z = 8$ (Oxygen). It has 8 protons ($+8$) and 10 electrons (-10), giving a net charge of -2 (O^{2-} anion).
Q.25	(A)	Carbon-14 decays with a half-life of 5,730 years, utilized to date ancient organic artifacts up to 50,000 years old.